



SARDAR PATEL UNIVERSITY

Bachelor of Computer Application (5th Semester NC) Examination

Code No:US05DBCA56 || Computer Graphics

Date : 08 - 10 - 2025 (Wednesday)

Time : 10:00 AM to 01:00 PM

Note : Right hand figure indicates the marks of each question.

Total Marks :[70]

Q.1 Multiple Choice Questions

[10]

- (1) _____ is not the application of Computer Graphics.
 (a) Computer Aided Design (b) Presentation Graphics
 (c) Power Point (d) Computer Art
- (2) CAD stands for _____.
 (a) Computer Aided Design (b) Common Application Design
 (c) Common Aided Design (d) Computer Application Design
- (3) CRT stands for _____.
 (a) Cathode Ray Tube (b) Computer Ray Tube
 (c) Common Ray Tube (d) None of the above
- (4) DDA stands for _____.
 (a) Data Drawing Algorithm (b) Digital Drawing Algorithm
 (c) Digital Differential Analyzer (d) Digital Drawing Analyzer
- (5) The overall design style for a set of characters is called _____.
 (a) Character Style (b) Face Style
 (c) Typeface (d) Character Face
- (6) Originally, _____ referred to a set of cast metal character forms.
 (a) Font (b) Character
 (c) Metal Words (d) Casted Words
- (7) Rotation point is also known as _____.
 (a) Rotation Angle (b) Rotation Axis
 (c) Pivot Point (d) Translation Vector
- (8) A scaling transformation alters the _____ of an object.
 (a) Size (b) Orientation
 (c) Shape (d) None of these
- (9) _____ mode application initiates data entry
 (a) Request Mode (b) Event Mode
 (c) Sample Mode (d) None of Above
- (10) _____ device used for selecting menu options
 (a) Locator Device (b) Choice Device
 (c) Valuator Device (d) Pick Device

Q.2 Short Questions [Attempt any Ten]

- (1) State disadvantages of DVST(Direct View Storage Tube).
- (2) List primary components of CRT and state its functions.
- (3) Explain the working of Joystick in short.
- (4) List attributes for the line. Explain any one
- (5) List various character attributes.
- (6) List caps used at the end of line. Explain any one
- (7) Define: 1. Reflection 2. Rotation
- (8) What is clipping? List out all clipping object
- (9) What is scaling? And list out its types
- (10) List various input modes.
- (11) List various interactive picture construction techniques.
- (12) What is a user model?

Q.3 [A] Explain working of CRT in detail.

[B] Write note on the Data Glove and Light pen.

[05]

OR

[A] Explain in detail raster scan.

[B] Write note on the Keyboard and Mouse.

[05]

[05]

Q.4 [A] Write steps of DDA Algorithm.

[B] Write a short note on character generation

[05]

OR

[A] Explain Area fill attributes.

[B] Explain odd-even rule with example.

[05]

[05]

Q.5 [A] Write a short note on Viewing Pipeline.

[B] Explain in brief Sutherland Hodgeman polygon clipping algorithm.

[05]

OR

[A] Explain window-to-viewport coordinate window transformation.

[B] Explain in brief Cohen Sutherland Line Clipping Algorithm.

[05]

[05]

Q.6 [A] List and explain logical classification of Input Devices in detail

[B] Explain Polygon surfaces in detail.

[05]

OR

[A] Explain in detail general considerations in structuring a user dialogue

[B] List and explain interactive picture-construction techniques

[05]

[05]

X

(2)